

Final – MATH 346/377 – Winter 2025

DATE: Tuesday, April 15 TIME: 9:00AM – 12:00PM

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- There are 11 problems, each worth 10 points. For Math 346, only your 8 best problems will count, for a total of 80 points. For Math 377, your 9 best problems will count, for a total of 90 points.
- Notes, calculators and other electronic devices are not permitted.

This page will not be graded and can be used as scrap paper.

1. (10 points) True or false? Justify your answer.
- (a) Let p be an odd prime. If g is a primitive root modulo p , then it is not a quadratic residue modulo p .
 - (b) $x = -\frac{1}{3}$ is a 3-adic *integer* (in $\mathbb{Z}_3$).
 - (c) Let p be a prime and a an integer. If $\gcd(a, p^2) = p$ then $\gcd(a + p, p^2) = p$.
 - (d) There exists an integer n for which the system $7x + (4n)y = 13$ has no solutions $x, y \in \mathbb{Z}$.

Solution:

- (a) TRUE. If g is a quadratic residue then $g \equiv r^2 \pmod{p}$, so that $g^{\frac{p-1}{2}} \equiv r^{p-1} \equiv 1 \pmod{p}$. Hence the order of g cannot be $p-1$. So
- (b) FALSE. The 3-adic norm of x is $|x|_3 = 3^{-v_3(x)} = 3 > 1$, so x is in $\mathbb{Q}_3$ but not in $\mathbb{Z}_3$.
- (c) FALSE. We can take $a = p(p-1) = p^2 - p$, for which $\gcd(a, p^2) = p$ but $\gcd(a + p, p^2) = p^2$.
- (d) TRUE. The equation $ax + by = c$ has a solution if and only if $\gcd(a, b)$ divides c . So if we take $n = 7$, then $\gcd(7, 28) = 7$ does not divide 13.

2. (10 points) Show that the equation

$$g(x) = (x^2 - 5)(x^2 - 41)(x^2 - 205)$$

has no integer solutions, but the congruence $g(x) \equiv 0 \pmod{p}$ has a solution for every prime p .

Solution: The polynomial has 6 real roots $\pm\sqrt{5}, \pm\sqrt{41}$ and $\pm\sqrt{205}$, which are not integers.

Modulo 2 the polynomial $g(x)$ has the root $x = 1$.

Modulo 5 the polynomial is $g(x) \equiv x^4(x^2 - 41) \pmod{5}$ which has the root $x = 0$.

Modulo 41 the polynomial is $g(x) \equiv x^4(x^2 - 5) \pmod{41}$ which also has the root $x = 0$.

Finally, for all the other primes $p \neq 2, 5, 41$ we have

$$\left(\frac{5}{p}\right) \left(\frac{41}{p}\right) = \left(\frac{205}{p}\right).$$

At least one of the three Legendre symbols must be 1, for which the corresponding quadratic polynomial has a root. Hence $g(x)$ also has a root modulo $p \neq 2, 5, 41$.

3. (10 points) Find all integers $0 \leq x \leq 24$ that satisfy

$$x^3 \equiv 13 \pmod{25}.$$

Solution: We start by solving modulo 5 and then lift the roots modulo 5^2 . Modulo 5 we have a unique solution $x_1 = 2$ that satisfies

$$x_1^2 = 8 \equiv 13 \pmod{5}.$$

(In particular, there is a unique solution modulo 5^2).

To lift it to a root x_2 modulo 5^2 , we set $x_2 = 2 + 5 \cdot k$ and determine the unique $0 \leq k \leq 4$ such that

$$13 \equiv x_2^3 \equiv (2 + 5 \cdot k)^3 \equiv 2^3 + 3 \cdot 2^2 \cdot 5k \equiv 8 + 60k \pmod{5^2}.$$

So 5^2 must divide $13 - 8 - 60k = 5 - 20k = 5(1 - 12k)$, which implies that $1 - 12k \equiv 1 - 2k \equiv 0 \pmod{5}$. We find that $k = 3$ and

$$x_2 = 2 + 5 \cdot 3 = 17.$$

Hence the unique integer $0 \leq x \leq 24$ that satisfies $x^3 \equiv 13 \pmod{25}$ is $x = 17$.

4. (a) (6 points) Find all residues of order 5 modulo 31.
 (b) (4 points) Deduce all integers $1 \leq x \leq 30$ solving

$$1 + x + x^2 + x^3 + x^4 \equiv 0 \pmod{31}.$$

Solution:

- (a) The integer 2 has order 5 since

$$2^5 \equiv 32 \equiv 1 \pmod{31}$$

and 5 is prime. The other elements of order 5 are of the form 2^i with i coprime to 5

$$x \equiv \{2, 2^2, 2^3, 2^4\} \pmod{31}.$$

So the four elements of order 5 are $\{2, 4, 8, 16\}$.

- (b) First, it is clear that $x = 1$ is not a solution since $1 + 1^2 + 1^3 + 1^4 = 4 \not\equiv 0 \pmod{31}$. Hence if x is a solution then we can multiply both sides by $x - 1 \not\equiv 0 \pmod{31}$ to get

$$0 \equiv (1 - x)(1 + x + x^2 + x^3 + x^4) \equiv 1 - x^5 \pmod{31}.$$

We deduce that the order of x modulo 31 must divide 5, so the order is 1 or 5. But $x \neq 1$, so in fact the order must be 5. We have established that the solutions to the equations are the elements of order 5 mod 31. From part a), it follows that the four solutions are $\{2, 4, 8, 16\}$.

5. (10 points) What are the last two digits of $(2357)^{3^{20}}$? Justify your answer.

Solution: The last two digits are given by the residue

$$(2357)^{3^{20}} \pmod{100}.$$

Note that $\phi(100) = 40$ and $\phi(40) = 16$. Since $20 \equiv 4 \pmod{16}$ and 3 is coprime to 40, it follows from Fermat's Little Theorem that

$$3^{20} = 3^4 \equiv 9^2 = 81 \equiv 1 \pmod{40}.$$

The integer 2357 is odd and not divisible by 5, hence is coprime to 100. Since $\phi(100) = 40$, it follows from FLT again that

$$(2357)^{3^{20}} \equiv 2357 \equiv 57 \pmod{100}.$$

Thus, the last two digits are 57.

6. (10 points) Find all integers $0 \leq x \leq 92$ that solve the equation

$$31x^{30} + 5x - 3 \equiv 0 \pmod{93 = 3 \cdot 31}.$$

Solution: By the Chinese Remainder Theorem, the integer x is a solution modulo 93 if and only if it is a solution modulo 3 and modulo 31.

Modulo 31, the equation reduces to

$$5x - 3 \equiv 0 \pmod{31},$$

which has a unique solution $x \equiv 3 \cdot 5^{-1} \pmod{31}$. Since $31 = 5 \cdot 6 + 1$, the inverse of 5 modulo 31 is -6 , and the solution to $5x - 3 \equiv 0 \pmod{31}$ is

$$x \equiv -18 \equiv 13 \pmod{31}.$$

On the other hand, modulo 3 the equation reduces to

$$x^{30} + 2x \equiv 0 \pmod{3}.$$

Then either $x \equiv 0 \pmod{3}$, or $x^{30} \equiv (x^2)^{15} \equiv 1 \pmod{3}$ by FLT. In the second case we find that $1 + 2x \equiv 0 \pmod{3}$, so the two solutions are $x \equiv 0, 1 \pmod{3}$.

Finally, we use the Chinese Remainder Theorem to find the root modulo 93. There are three integers $0 \leq x \leq 92$ congruent to 2 modulo 31:

$$\{13, 44, 75\}.$$

We have $13 \equiv 1 \pmod{3}$, $44 \equiv 2 \pmod{3}$ and $75 \equiv 0 \pmod{3}$. Hence we have the two solutions

$$\{13, 75\}$$

to $31x^{30} + 5x - 3 \equiv 0 \pmod{93 = 3 \cdot 31}$.

7. (a) (5 points) For which primes p is $p^2 + 1$ prime?
(b) (5 points) For which primes p is $p^2 + 2$ prime?

Solution:

- (a) If p is an odd prime, then p^2 is odd and $p^2 + 1$ is not a prime since it is even. For $p = 2$ we have $p^2 + 1 = 5$ is prime. So $p^2 + 1$ is prime if and only if $p = 2$.
- (b) Similarly, if p is not congruent to 3 modulo, then p^2 is always congruent to 1 modulo. So $p^2 + 2 \equiv 0 \pmod{3}$ and not equal to 3, since $p^2 + 2 > 3$. On the other hand, for $p = 3$ we have $p^2 + 2 = 9 + 2 = 11$ is prime. So $p = 3$ is the only prime for which $p^2 + 2$ is prime.

8. (10 points) Use Euclid's idea to prove (by contradiction) that there are infinitely many primes congruent to 3 modulo 4.

Solution: The proof is similar to the proof that there are infinitely many primes congruent to 2 modulo 3. First, note that any prime $p > 2$ is congruent to 1 or 3 modulo 4. Moreover, we will also need that the product of two primes congruent to 1 modulo 4 is also congruent to 1 modulo 4.

Suppose that there are only finitely many primes $S = \{3, p_1, \dots, p_r\}$ that are congruent to 3 modulo 4. Consider $n = 4 \cdot p_1 \cdots p_r + 3$ (we exclude 3 from the product). Since the primes p_i are coprime to 3, n is coprime to 3 and to all the other primes $p_i \in S$. Hence n is only divisible by primes $q \notin S$, that are congruent to 1 modulo 4. But if all primes dividing n are congruent to 1 modulo 4, n must be congruent to 1 modulo 4 which is not the case. This contradicts the finiteness of S .

9. (a) (5 points) Let g be a primitive root modulo a prime $p \equiv 1 \pmod{4}$. Show that $g^{\frac{p-1}{4}}$ is a root of $x^2 + 1 \equiv 0 \pmod{p}$.
- (b) (5 points) Use part (a) to find the two roots of $x^2 + 1 \equiv 0 \pmod{13}$.

Solution:

- (a) Let us set $k = g^{\frac{p-1}{4}}$. Then $k^4 = g^{p-1} \equiv 1 \pmod{p}$, so k is a root of

$$x^4 - 1 = (x^2 - 1)(x^2 + 1) \equiv 0 \pmod{p}.$$

On the other hand, $k^2 = g^{\frac{p-1}{2}} \not\equiv 1 \pmod{p}$, since g is a primitive root and it has order $p-1$. So k must be a root of $x^2 + 1 \equiv 0 \pmod{p}$.

- (b) Modulo $p = 13$ we can check that $g = 2$ is a primitive root since

$$\begin{aligned} 2^{12} &\equiv 1 \pmod{13}, \\ 2^{12/2} &\equiv 2^6 \equiv 64 \equiv -1 \not\equiv 1 \pmod{13}, \\ 2^{12/3} &\equiv 2^4 \equiv 16 \equiv 3 \not\equiv 1 \pmod{13}. \end{aligned}$$

By part a), we find that $k = 2^{12/4} = 2^3 = 8$ satisfies $x^2 + 1 \equiv 0 \pmod{13}$. The other root is $k = 13 - 8 = 5$. (Indeed $8^2 + 1 \equiv 65 \equiv 0 \pmod{13}$ and $5^2 + 1 \equiv 26 \equiv 0 \pmod{13}$).

10. (a) (8 points) Show that the Euler ϕ -function satisfies $\phi(n^2) = n\phi(n)$ for any integer $n \geq 1$. Deduce that for $k \geq 2$ and p a prime

$$\phi(\phi(p^k)) = p^{k-2}\phi((p-1)^2).$$

- (b) (2 points) Compute $\phi(\phi(5^3))$.

Solution:

- (a) If $n = 1$ the result is immediate. So assume $n > 1$ and let $n = \prod_{i=1}^r p_i^{e_i}$ be the prime factorization of n , where $e_i \geq 1$. Then $n^2 = \prod_{i=1}^r p_i^{2e_i}$ and

$$\begin{aligned}\phi(n^2) &= \prod_{i=1}^r \phi(p_i^{2e_i}) = \prod_{i=1}^r p_i^{2e_i-1}(p_i - 1) \\ &= \left(\prod_{i=1}^r p_i^{e_i} \right) \left(\prod_{i=1}^r p_i^{e_i-1}(p_i - 1) \right) \\ &= n\phi(n).\end{aligned}$$

We deduce that

$$\begin{aligned}\phi(\phi(p^k)) &= \phi(p^{k-1}(p-1)) = \phi(p^{k-1})\phi(p-1) \\ &= p^{k-2}(p-1)\phi(p-1) \\ &= p^{k-2}\phi((p-1)^2).\end{aligned}$$

- (b) Using the formula we find that

$$\phi(\phi(5^3)) = 5\phi(4^2) = 5 \cdot 8 = 40.$$

11. For an odd prime p we consider the sum

$$S(p) := \sum_{k=1}^{p-1} \left(\frac{k}{p} \right).$$

- (a) (4 points) For $p = 5$, verify that $S(5) = 0$.
(b) (6 points) Show that $S(p) = 0$ for all odd primes.

Solution:

- (a) (4 points) For $p = 5$, the quadratic residues are 1, 4, and the non-residues are 2, 3. Hence, we have

$$S(5) = \left(\frac{1}{5} \right) + \left(\frac{2}{5} \right) + \left(\frac{3}{5} \right) + \left(\frac{4}{5} \right) = 1 + (-1) + (-1) + 1 = 0. \quad (1)$$

- (b) (6 points) Now let p be any odd prime. Let g be a primitive root modulo p . Each residue $1 \leq k \leq p-1$ is congruent to a unique $g^j \pmod{p}$. Hence, we can rewrite the sum as

$$S(p) = \sum_{j=0}^{p-2} \left(\frac{g^j}{p} \right).$$

We have seen in class that

$$\left(\frac{g^j}{p} \right) = (-1)^j,$$

so the sum is equal to

$$\sum_{j=0}^{p-2} (-1)^j = 1 + (-1) + 1 + (-1) + \cdots + 1 + (-1) = 0.$$

